



LASU Certificate Generator

OOP Design Patterns · C++17 · Winsock HTTP Server · Web UI

CSC 226 — Object-Oriented Programming I — Lagos State University — July 2026

Strategy

TemplateMethod

Factory

Inheritance

Polymorphism

RAII

std::map

Multithreading

Group 6: Idowu Fawaz, Idowu Matthew, Ikechukwu Queen, Inedu Esther, Iniotuh Greatest, Iwakun Oluwasegun, Iwuno Vincent, Iyaniwura George, Jawando Fuad, Jeremiah David, Joseph Elizabeth, Kalejaiye Halimah, Kasali Damilola, Kassim Oluwatimilehin, Kehinde Oyindamola, Kolawole Abubakar, Kwegan Sean, Lamina Rihanat, Lawal Sahal, Lawal Ibrahim

Overview & Architecture

Desktop webapp: generate, manage, and print HTML certificates

3 CertificateTypes

Excellence, Completion, Participation

3 Template Styles

Formal(serif),Modern(gradient), Minimalist (mono)

CSV Batch Import

File path or paste · 10 sample students

Search, Sort & Stats

Name,course,GPA · Per-course & grade distro



Data flow: Browser→HTTP(JSON) → HttpServer → CertificateGenerator → Certificate + Template → HTML file

Inheritance & Encapsulation

Person (abstract) → Student (concrete) · All data private

```
class Person { // Abstractbase
protected:
    std::string name_;
public:
    virtual ~Person() = default;
    virtual std::string
        getDescription()const= 0; // pure virtual
};

class Student : public Person { // is-a Person
    std::string course_, grade_;
public:
    double gradeToPoints() const;
    // 'A' ->4.0 'B+' ->3.3 '93' ->3.72
    static bool isValidGrade(const&);
    std::string getDescription()
        const override;
};
```

```
// Encapsulation in Generator
class CertificateGenerator {
    std::map<std::string, Student>
        students_; // private
    mutable std::mutex mutex_; // private
public:
    void addStudent(const&); // validated
    bool removeStudent(const&);
    Student* findStudent(const&);
    void loadFromCsv(const&);
};
```

- **Inheritance:** Student is-a Person — inherits name_, adds course/grade
- **Encapsulation:** All data private — addStudent() validates before storing, throws on bad data

Polymorphism & Abstract Base Classes

3 ABCs, 7 pure virtuals, runtime vtable dispatch

```
class Certificate{           // Abstractbase
public:
    virtual std::string
        generateHTML()const;    // Template Method
    virtual std::string
        getTypeName()const= 0; // pure virtual
};

class CertOfExcellence : public Certificate {
    std::string getTypeName()constoverride
    { return"Excellence"; }
};

class CertOfCompletion : public Certificate {
    std::string getTypeName()constoverride
    { return"Completion"; }
};
```

- **3 ABCs:** Person, CertificateTemplate, Certificate
- **7 pure virtual methods** total
- Runtime dispatch via vtable
- Client code works with base pointers

```
// Factory return subclass at runtime
auto tpl = makeTemplate(style);
auto cert = makeCertificate(
    type, s, std::move(tpl));
file << cert->generateHTML(); // polymorphic!
```

Strategy + Template Method

Strategy: swappable visual styles · Template Method: fixed skeleton

```
// STRATEGY: CertificateTemplate
class CertificateTemplate {
public:
    virtual string header(...)const= 0;
    virtual string styles()const= 0;
    virtual string body(...)const= 0;
};
class FormalTemplate : public CertTempl {
    // Double border, serif (Georgia)
};
class ModernTemplate : public CertTempl {
    // Gradient bg, rounded corners
};
class MinimalistTemplate : public CertTempl {
    // Monospace font, clean lines
};
```

```
// TEMPLATE METHOD: generateHTML()
string Certificate::generateHTML()const{
    string typeName = getTypeName(); // virtual
    string html;
    html += template_.header(typeName);
    html += template_.styles();
    html += template_.body(student_,
                           typeName, date);

    html += "</body></html>";
    return html;
}
// Subclass provides type name/desc
// Template provides all HTML/CSS
```

- **Strategy:** Open/Closed — new templates don't change Certificate | Style chosen at runtime
- **Template Method:** getTypeName() dispatches polymorphically | Zero duplication across types

Factory + Composition

makeTemplate/makeCertificate· has-a relationships throughout

```
//Factory: returns correct subclass
unique_ptr<CertificateTemplate>
makeTemplate(TemplateStyle style){
    switch (style) {
        case Formal:
            return mk<FormalTemplate>();
        case Modern:
            return mk<ModernTemplate>();
        case Minimalist:
            return mk<MinimalistTemplate>();
    }
}
```

```
// Composition: has-a
class Certificate {
    const Student& student_;
    unique_ptr<CertificateTemplate>
        template_; // has-a Template
};
```

- **Factory:** Specify an enum → get right polymorphic type
- No conditional logic at call site
- Add type: enum value + case + class
- **Composition:** has-a beats is-a
- Runtime flexibility (swap template)
- Looser coupling, easier testing
- HttpServer has-a Generator
- Generator has-a map of Students

RAII & STL Features

Automatic resource management · std::map, sort, find_if, lambdas, thread

```
// RAII: no manual cleanup
void addStudent(const Student& s) {
    std::lock_guard<mutex> lock(m_);
    students_[toLower(s.getName())] = s;
    // auto-unlocked at scope exit
}

// unique_ptr: auto-deleted polymorphic
auto cert = makeCertificate(type, s, t);
// cert + tmp auto-deleted when done
```

```
//STL: case-insensitive search
Student* findStudent(const string&n){
    auto it = find_if(begin(students_),
        end(students_),
        [&](const auto& p) {
            return toLower(p.first)
                == toLower(n);
        });
    return it != end(students_)
        ? &it->second : nullptr;
}
```

- **RAII:** unique_ptr, lock_guard, ofstream — zero manual delete/unlock/close
- **STL:** std::map (O(log n)), find_if+lambda, sort+lambda, thread, mutex, stringstream

HTTP Server & REST API

WinsockTCP · 12endpoints · Embedded SPA · Thread safety

```
// Accept loop: each conn on its own thread
while (true) {
    SOCKET client = accept(server,
        nullptr, nullptr);
    if (client == INVALID_SOCKET) break;
    std::thread(handleClient, client,
        std::ref(gen)).detach();
}
```

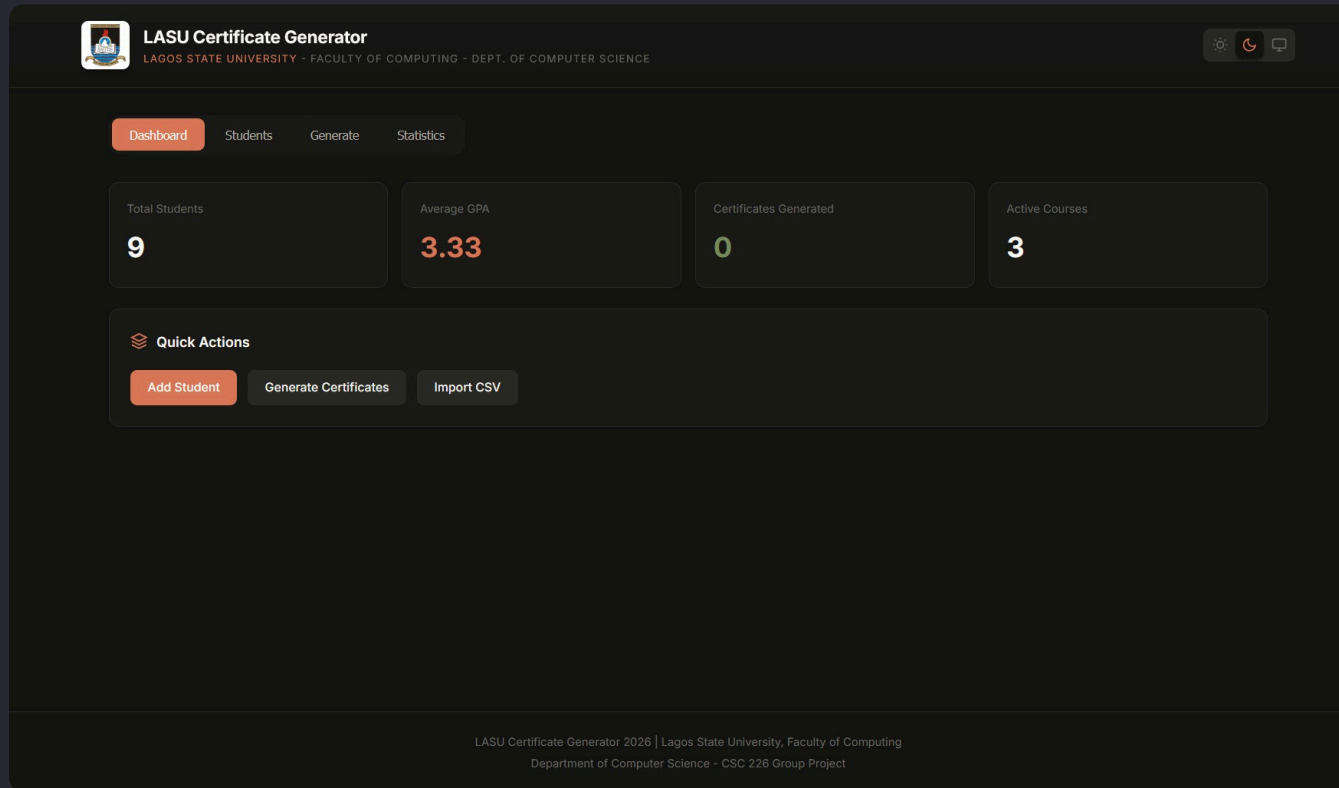
- All HTML/CSS/JS in C++ raw string literal
- Frontend uses fetch() — no reloads
- Theme toggle: Light/Dark/System

Method	Path	Action
GET	/	Serve Web UI
GET	/list	All students JSON
GET	/stats	Statistics JSON
POST	/add	Insert student
POST	/remove	Delete student
POST	/load	CSV import
POST	/generate	Create certificates
POST	/sort	Sort students

- std::mutex guards map (thread-safe)
- ValidationException → JSON error
- CSV: invalid lines skipped

Web UI Preview

Browser-based interface — Dashboard, Students, Generate, Statistics tabs



- **Dashboard:** Total students, avg GPA, cert count, active courses
- **Students:** Search, sort, add, remove — grade badges with colors
- **Generate:** Pick type + template style — one student or all
- **Statistics:** Course distribution, grade distribution
- **Theme toggle:** Light / Dark / Match System

All Concepts in Action

End-to-end: add student → generate certificate → HTML output

```
// 1. Add (encapsulation try { +exception)

    gen.addStudent(
        Student("Alice", "CSC 226", "A"));
} catch (const ValidationE& e) {
    cerr << e.what();
}

// 2. Create Template (factory + strategy) auto tmp1
= makeTemplate(Formal);

// 3. Create Certificate (factory + poly.) auto cert
= makeCertificate(
    Excellence, student, move(tmp1));
```

```
// 4. Generate HTML (templatemethod)
string html = cert->generateHTML();
//     getTypeName() → "Excellence"
//     FormalTemplate::header() + styles()
//     FormalTemplate::body()

// 5. Write file (RAII: ofstream) ofstream
file("output/Alice.html"); file << html; //
auto-closed

// 6. Serve via HTTP (thread + mutex) //
Browser GET /list returns JSON
```

Inheritance

Student : Person

Polymorphism

cert->generateHTML()

Encapsulation

private + getters

Abstract Base

3 ABCs, 7 virtuals

Strategy

FormalTemplate CSS

Template Method

generateHTML()

Factory

makeTemplate/Cert

Composition

has-a Template

RAII

unique_ptr/lock

STL

map/sort/thread

Exception

try/catch validate

Thread Safety

mutex guard

Thank You

Lagos State University—Department of Computer Science — CSC 226 — Group 6

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```
g++ -std=c++17 -I include main.cpp src/*.cpp -lws2_32 && ./certificate_generator
```

CONTRIBUTIONS

GROUP LEADERS : Iwuno Vincent ChukwuEbuka 240591107

GROUP MEMBERS

1. Idowu Fawaz Olawunmi 240591101
2. Idowu Matthew 240591102
3. Ikechukwu Queen Chisom 240591103
4. Inedu Esther Ogaicha 240591104
5. Iniotuh Greatest Akanimoh 240591105
6. Iwakun Oluwasegun 240591106
7. Iwuno Vincent ChukwuEbuka 240591107
8. Iyaniwura Olabamiji George 240591108
9. Jawando Fuad Olamide 240591109
10. Jeremiah David Preye 240591110
11. Joseph Elizabeth Nifemi 240591111
12. Kalejaiye Halimah Temilade 240591112
13. Kasali Damilola 240591113
14. Kassim Oluwatimilehin A. 240591114
15. Kehinde Oyindamola Ayomide 240591115
16. Kolawole Abubakar Olaoluwa 240591116
17. Kwegan Sean Oluwatomilade 240591117
18. Lamina Rihanat Opemipo 240591118
19. Lawal Sahal Adeshayo 240591119
20. Lawal Ibrahim Oluwafemi 240591120